

PERTURBATION THEORY: A SUPER-FAST INTRODUCTION

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The following is really super condensate (although self-consistent). If you want more details see [1] in which you probably can find more than you are looking for.

1. FUNCTIONAL CALCULUS

First of all notice that all the Riemannian theory of integration works verbatim¹ for function $f \in \mathcal{C}^0(\mathbb{R}, \mathbb{B})$, where $\mathbb{B}$ is a Banach space.² We can thus talk of integrals of the type $\int_a^b f(t)dt$, it is special case of the so called Bochner integral. Next, we can talk of analytic functions for functions in $\mathcal{C}^0(\mathbb{C}, \mathbb{B})$: a function is analytic in an open region $U \subset \mathbb{C}$ iff at each point $z_0 \in U$ there exists a neighborhood $B \ni z_0$ and elements $\{a_n\} \subset \mathbb{B}$ such that

$$(1.1) \quad f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \forall z \in B.$$

Problem 1. Show that if $f \in \mathcal{C}^0(\mathbb{C}, \mathbb{B})$ is analytic in $U \subset \mathbb{C}$, then given any smooth closed curve γ , contained in a sufficiently small disk in U , holds³

$$(1.2) \quad \int_{\gamma} f(z) dz = 0$$

Then show that the same hold for any piecewise smooth close curve in U . (Hint: check that the same argument used in the well known case $\mathbb{B} = \mathbb{C}$ works also here.)

Problem 2. Show that if $f \in \mathcal{C}^0(\mathbb{C}, \mathbb{B})$ is analytic in $U \subset \mathbb{C}$, then given any smooth closed curve γ , contained in U and having in its interior a point z , holds the formula

$$(1.3) \quad f(z) = \frac{1}{2\pi i} \int_{\gamma} (\xi - z)^{-1} f(\xi) d\xi.$$

(Hint: check that the same argument used in the well known case $\mathbb{B} = \mathbb{C}$ works also here.)

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¹Of course one cannot talk of sup or inf of the sums, yet one can easily show that any two sufficiently fine partitions yields close values, just take the partition given by the intersection and compare with that.

²Recall that a Banach space is a complete normed vector space (in the following we will consider only vector spaces on the field of complex numbers), that is a normed vector space in which all the Cauchy sequences have a limit in the space.

³Of course, by $\int_{\gamma} f(z) dz$ we mean that we have to consider any smooth parametrization $g : [a, b] \rightarrow \mathbb{C}$ of γ , $g(a) = g(b)$, and then $\int_{\gamma} f(z) dz := \int_a^b f \circ g(t) g'(t) dt$. Show that the definition does not depend on the parametrization and that one can use piecewise smooth parametrizations as well.

Problem 3. Show that if $f \in C^0(\mathbb{C}, \mathbb{B})$ satisfies (1.3) for each smooth closed curve in an open set U , then f is analytic in U . (Hint: check that the same argument used in the well known case $\mathbb{B} = \mathbb{C}$ works also here.)

Problem 4. Show that the $n \times n$ matrices form a Banach Algebra.⁴ (Hint: use the norm $\|A\| = \sup_{v \in \mathbb{R}^n} \frac{\|Av\|}{\|v\|}$.)

Problem 5. Show that the set of bounded linear operators from a Banach space $\mathbb{B}$ to itself (usually called $L(\mathbb{B}, \mathbb{B})$) form a Banach algebra. (Hint: Use the same norm as in Problem 4.)

2. SPECTRUM AND RESOLVENT

Given $A \in L(\mathbb{B}, \mathbb{B})$ we define the resolvent, called $\rho(A)$, as the set of the $z \in \mathbb{C}$ such that $(z\mathbf{Id} - A)$ is invertible and the inverse belongs to $L(\mathbb{B}, \mathbb{B})$. The spectrum of A , called $\sigma(A)$ is the complement of $\rho(A)$ in $\mathbb{C}$.

Problem 6. Prove that, for each Banach space $\mathbb{B}$ and operator $A \in L(\mathbb{B}, \mathbb{B})$, if $z \in \rho(A)$, then there exists a neighborhood U of z such that $(z\mathbf{Id} - A)^{-1}$ is analytic in U . (Hint: Note that

$$(\zeta\mathbf{Id} - A) = (z\mathbf{Id} - A - (z - \zeta)\mathbf{Id}) = (z\mathbf{Id} - A) [\mathbf{Id} - (z - \zeta)(z\mathbf{Id} - A)^{-1}]$$

and that if $\|(z - \zeta)(z\mathbf{Id} - A)^{-1}\| < 1$ then the inverse of $\mathbf{Id} - (z - \zeta)(z\mathbf{Id} - A)^{-1}$ is given by $\sum_{n=0}^{\infty} (z - \zeta)^n [(z\mathbf{Id} - A)^{-1}]^n$ (the Von Neumann series—which really is just the geometric series).)

From the above exercise follows that $\rho(A)$ is open, hence $\sigma(A)$ is closed.

Problem 7. Show that, for each $A \in L(\mathbb{B}, \mathbb{B})$, $\sigma(A) \neq \emptyset$. (Hint: If $\sigma(A) = \emptyset$, then $(z\mathbf{Id} - A)^{-1}$ is an entire function, then the Von Neumann series shows that $(z\mathbf{Id} - A)^{-1} = z^{-1}(\mathbf{Id} - z^{-1}A)^{-1}$ goes to zero for large z , and then (1.3) shows that $(z\mathbf{Id} - A)^{-1} = 0$ which is impossible.)

Next, consider a Banach space $\mathbb{B}$ and $A \in L(\mathbb{B}, \mathbb{B})$, given a smooth curve γ containing in its interior $\sigma(A) \cup \{0\}$, for each $n \in \mathbb{N}$ holds

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} (z\mathbf{Id} - A)^{-1} dz &= \frac{1}{2\pi i} \int_{\gamma} z^{-1} (\mathbf{Id} - z^{-1}A)^{-1} dz \\ &= \lim_{r \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} (\mathbf{Id} - r^{-1}e^{-i\theta}A)^{-1} d\theta = \mathbf{Id}. \end{aligned}$$

Problem 8. Show that, if $A \in L(\mathbb{B}, \mathbb{B})$ and p is any polynomial, then for each $n \in \mathbb{N}$ and smooth curve $\gamma \subset \mathbb{C}$, with $\sigma(A)$ in its interior,

$$p(A) = \frac{1}{2\pi i} \int_{\gamma} p(z)(z\mathbf{Id} - A)^{-1} dz.$$

(Hint: from the above equality follows

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma} z^n (z\mathbf{Id} - A)^{-1} dz &= A^n + \frac{1}{2\pi i} \int_{\gamma} (z^n - A^n)(z\mathbf{Id} - A)^{-1} dz \\ &= A^n + \sum_{k=0}^{n-1} \frac{1}{2\pi i} \int_{\gamma} z^k A^{n-k} dz = A^n. \end{aligned}$$

⁴A Banach Algebra $\mathbb{A}$ is a Banach space where it is defined the multiplications between element with the usual properties of an algebra and, in addition, for each $a, b \in \mathbb{A}$ holds $\|ab\| \leq \|a\| \cdot \|b\|$.

The statement for polynomial follows trivially.)

Note that if $f(z) = \sum_{n=0}^{\infty} f_n z^n$ is an analytic function in all $\mathbb{C}$ (entire), then formula (1.1) allows to define

$$f(A) = \sum_{n=0}^{\infty} f_n A^n$$

Problem 9. Show that, if $A \in L(\mathbb{B}, \mathbb{B})$ and f is an entire function, then for each smooth curve $\gamma \subset \mathbb{C}$, with $\sigma(A)$ in its interior,

$$f(A) = \frac{1}{2\pi i} \int_{\gamma} f(z)(z\mathbf{Id} - A)^{-1} dz.$$

(Hint: approximate with polynomials.)

In view of the above fact, the following definition is natural:

Definition 1. For each $A \in L(\mathbb{B}, \mathbb{B})$, f analytic in a region U containing $\sigma(A)$, then for each smooth curve $\gamma \subset U$, with $\sigma(A)$ in its interior, define

$$f(A) = \frac{1}{2\pi i} \int_{\gamma} f(z)(z\mathbf{Id} - A)^{-1} dz.$$

Problem 10. Show that the above definition does not depend on the curve γ .

Problem 11. Suppose that $A \in L(\mathbb{B}, \mathbb{B})$ and $\sigma(A) = B \cup C$, $B \cap C = \emptyset$, suppose that the smooth closed curve $\gamma \subset \rho(A)$ contains B in its interior, prove that

$$P_B := \frac{1}{2\pi i} \int_{\gamma} (z\mathbf{Id} - A)^{-1} dz$$

is a projector that does not depend on γ . *(Hint: The non dependence on γ is obvious. A projector is characterized by the property $P^2 = P$. Thus*

$$\begin{aligned} P_B^2 &:= \frac{1}{(2\pi i)^2} \int_{\gamma_1} \int_{\gamma_2} (z\mathbf{Id} - A)^{-1} (\zeta\mathbf{Id} - A)^{-1} dz d\zeta \\ &= \frac{1}{(2\pi i)^2} \int_{\gamma_1} dz \int_{\gamma_2} d\zeta (z - \zeta)^{-1} [(z\mathbf{Id} - A)^{-1} - (\zeta\mathbf{Id} - A)^{-1}]. \end{aligned}$$

If we have chosen γ_1 in the interior of γ_2 , then $(z - \zeta)^{-1}(\zeta\mathbf{Id} - A)^{-1}$ is analytic in the interior of γ_1 , hence the corresponding integral gives zero. The other integral gives P_B , as announced.)

Problem 12. Prove that, if $P \in L(\mathbb{B}, \mathbb{B})$ and $P^2 = P$, then setting $R(P) = P\mathbb{B}$ and $N(P) = \{v \in \mathbb{B} : Pv = 0\}$ holds $\mathbb{B} = R(P) \oplus N(P)$. *(Hint: For each $v \in \mathbb{B}$, write $v = Pv + (\mathbf{Id} - P)v$.)*

Note that from the definition in Problem 11 easily follows that $P_B A = A P_B$. Hence, $AR(P_B) \subset R(P_B)$ and $AN(P_B) \subset N(P_B)$. Thus $\mathbb{B} = R(P_B) \oplus N(P_B)$ provides an invariant decomposition for A .

Problem 13. Prove that $A = P_B A P_B + (\mathbf{Id} - P_B) A (\mathbf{Id} - P_B)$. *(Hint: use the above decomposition and the fact that $(\mathbf{Id} - P_B)$ is a projector.)*

Problem 14. Prove that $\sigma(P_B A P_B) = B \cup \{0\}$. Moreover, if $\dim(R(P_B)) = D < \infty$, then the cardinality of B is $\leq D$. (Hint: the first part follows from the previous decomposition, the second from the fact that $P_B A P_B$, when restricted to the space $R(P_B)$ is described by a $D \times D$ matrix A_B and the equation $\det(z \mathbf{Id} - A_B) = 0$ is a polynomial of degree D in z and hence has exactly D solutions (counted with multiplicity).⁵).

3. PERTURBATIONS

Let us consider $A, B \in L(\mathbb{B}, \mathbb{B})$ and the family of operators $A_\nu := A + \nu B$.

Lemma 3.1. For each $\delta > 0$ there exists $\nu_\delta \in \mathbb{R}$ such that, for all $|\nu| \leq \nu_\delta$, $\rho(A_\nu) \supset \{z \in \mathbb{C} : d(z, \sigma(A)) > \delta\}$.

Proof. Let $d(z, \sigma(A)) > \delta$, then

$$(3.1) \quad (z \mathbf{Id} - A_\nu) = (z \mathbf{Id} - A) [\mathbf{Id} - \nu(z \mathbf{Id} - A)^{-1} B]$$

Now $\|(z \mathbf{Id} - A)^{-1} B\|$ is a continuous function in z outside $\sigma(A)$, moreover it is bounded outside a ball of large enough radius, hence there exists $M_\delta > 0$ such that $\sum_{d(z, \sigma(A)) > \delta} \|(z \mathbf{Id} - A)^{-1} B\| \leq M_\delta$. Choosing $\nu_\delta = (2M_\delta)^{-1}$ yields the result. $\square$

Suppose that $\bar{z} \in \mathbb{C}$ is an isolated point of $\sigma(A)$, that is there exists $\delta > 0$ such that $\{z \in \mathbb{C} : |z - \bar{z}| \leq \delta\} \cap (\sigma(A) \setminus \{\bar{z}\}) = \emptyset$, then the above Lemma shows that, for ν small enough, $\{z \in \mathbb{C} : |z - \bar{z}| \leq \delta\}$ still contains an isolated part of the spectrum of $\sigma(A_\nu)$, let us call it B_ν , clearly $B_0 = \{\bar{z}\}$.

Problem 15. Let P_{B_ν} be defined as in Problem 11. Prove that, for ν small enough, it is an analytic function of ν . (Hint: Use the representation in Problem 11 and formula (3.1).)

Problem 16. If P, Q are two projectors and $\|P - Q\| < 1$, then $\dim(R(P)) = \dim(R(Q))$. (Hint: Note that $Q(\mathbf{Id} + P - Q) = QP$, then $Q = (\mathbf{Id} - (Q - P))^{-1} QP$, hence $\dim(R(P)) \geq \dim(R(Q))$, exchanging the role of P and Q the result follows.)

The above two exercises imply that the dimension of the eigenspace $R(P_\nu)$ is constant.

Lemma 3.2. If $\dim(R(P_0)) = 1$, then A_ν has a unique eigenvalue z_ν in a neighborhood of $\bar{z}$, $z_0 = \bar{z}$. In addition z_ν is an analytic function of ν .

Proof. From the previous exercises it follows that P_ν is a rank one operator which depend analytically on ν . Let $v \in R(P_0)$, then $Av = AP_0v = P_0Av$ hence $Av \in R(P_0)$, hence parallel to v . From this follows that there exists $\lambda \in \mathbb{R}$ such that $Av = \lambda v$, but then $\lambda \in \sigma(A)$. Moreover, $P_0AP_0v = P_0Av = \lambda P_0v = \lambda v$, thus $\lambda \in \sigma(P_0AP_0) = \{\bar{z}\}$. In addition, since P is a rank one projector it must have

⁵This is the real reason why spectral theory is done over the complex rather than the real. You should be well aquatinted with the fact that a polynomial p of degree D has D root over $\mathbb{C}$ but, in case you have forgotten, consider the following: first a polynomial of degree larger than zero must have at least a root, otherwise $\frac{1}{p(z)}$ would be an entire function and hence

$$\frac{1}{p(z)} = \lim_{r \rightarrow \infty} \frac{1}{2\pi} \int_0^{2\pi} d\theta \frac{1}{p(z + re^{i\theta})} = 0.$$

Let z_1 be a root. By the Taylor expansion in z_1 follows the decomposition $p(z) = (z - z_1)p_1(z)$ where p_1 has degree $D - 1$. The result follows by induction.

the form $Pw = v\ell(w)$, where $w \in \mathbb{B}'$.⁶ Next, let $v_\nu := (\ell(P_\nu v))^{-1}P_\nu v$, by the same argument $A_\nu v_\nu = z_\nu v_\nu$. Then $z_\nu = \ell(A_\nu v_\nu)$, since one can easily show that the right hand side is analytic, so must be the left hand side. $\square$

Hence in the case of $A \in L(\mathbb{B}, \mathbb{B})$ with an isolated simple⁷ eigenvalue $\bar{z}$ we have that the corresponding eigenvalue z_ν of $A_\nu = A + \nu B$, $B \in L(\mathbb{B}, \mathbb{B})$, for ν small enough, depend smoothly from ν . In addition, using the notation of the previous Lemma, we can easily compute the derivative: differentiating $A_\nu v_n u = z_\nu v_\nu$ with respect to ν and then setting $\nu = 0$, yields

$$Bv + Av'_0 = z'_0 v + \bar{z}v'_0.$$

But $Pv_n u = v\ell(v_\nu) = v$, thus $Pv'_\nu = 0$ and $PAv'_\nu = APv'_\nu = 0$, which implies

$$z'_0 = \ell(Bv).$$

Problem 17. Compute v'_0 .

Problem 18. What does it happen if the eigenspace associated to $\bar{z}$ is finite dimensional, but with dimension strictly larger than one? (Hint: think hard.⁸)

REFERENCES

- [1] T.Kato, *Perturbation theory for linear operators*, Springer (1966).

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⁶By $\mathbb{B}'$, the dual space, we mean the set of bounded linear functionals on $\mathbb{B}$. Verify that is a Banach space with the norm $\|\ell\| = \sum_{w \in \mathbb{B}} \frac{|\ell(w)|}{\|w\|}$.

⁷That is with the associated eigenprojector of rank one.

⁸A good idea is to start by considering concrete examples, for instance

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \mu \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + \mu \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$